

UNG FACILITIES PLANNING & GIS

Version current

February / March 2026

Tracks personnel movement between AD export dates — hires, departures, and office changes

AD_Delta_Comparison.py

Purpose

Compares two AD exports — a fixed baseline and the current file — and reports what changed between them. Captures new hires, departures, and field-level changes with particular focus on department and office location changes, which are the two fields that affect GIS matching. Produces evidence that the pipeline is capturing real institutional movement automatically.

When to Run

- Weekly, as Step 3 of the pipeline
- The baseline file is fixed at a meaningful reference point (currently March 19, 2026)
- Update BASELINE_FILE only when a new permanent reference point is established

Requirements

- Python with pandas (no ArcGIS required)
- Baseline AD Excel file
- Current raw AD Excel file (not the parsed version)

Inputs

Item	Value	Notes
Baseline file	FacStaffWithAddress_03192026 BASELINE.xlsx	Fixed reference point — update path only when baseline changes
New file	FacStaffWithAddress_MMDDYY YY.xlsx	Current raw AD export — update for each run

What It Does

- Loads both files with dtype=str to prevent phone number float truncation
- Normalizes email addresses to lowercase for reliable key matching
- Finds ADDS: records in new file but not baseline (new hires)
- Finds DELETES: records in baseline but not new file (departures)
- Finds CHANGES: field-level differences for records present in both files
- Produces a focused Key Changes sheet showing only department and office location changes — the two fields that matter for GIS matching

Outputs

Item	Value	Notes
AD_Delta_YYYYMMDD_HHMMSS.xlsx	baseline\DeltaReports\	Multi-sheet workbook
Summary sheet	In workbook	Counts at a glance including per-field key change breakdown
Adds sheet	In workbook	New employees since baseline
Deletes sheet	In workbook	Departed employees since baseline
Key_Changes sheet	In workbook	Department and office location changes only
Changes sheet	In workbook	All field-level changes, one row per change
Baseline_Reference sheet	In workbook	Full baseline for reference

Key Configuration

BASELINE_FILE: Fixed path to the baseline export — only change when establishing a new baseline

NEW_FILE: Update to current run filename each week

EXCLUDE_FROM_COMPARISON: Value.postaladdress excluded by default; uncomment telephone to exclude phone changes

KEY_CHANGE_FIELDS: Value.department and Value.physicaldeliveryofficename — the two GIS-critical fields